March 13, 2025

LEXSET

Office of Science and Technology Policy
National Sciences Foundation
1650 Pennsylvania Avenue
Washington, DC 20504

Comment on Request for Information on the Development of an AI Action Plan

Lexset welcomes the opportunity to respond to the National Science Foundation's Request for Information (RFI) on a National AI Action Plan. As a leading developer of synthetic data and systems for artificial intelligence, Lexset supports the Trump Administration's focus on sustaining American AI leadership while removing unnecessary regulatory barriers, as embodied in Executive Order 14179, *"Removing Barriers to American Leadership in Artificial Intelligence."* Our comments align closely with the administration's directive to enhance America's AI dominance in support of economic competitiveness and national security, and to ensure that overly burdensome requirements do not hamper private-sector innovation. We provide evidence of synthetic data's effectiveness and cost benefits, discuss its importance in the U.S.-China technological competition, describe Lexset's contributions through public-private partnerships, and recommend policies to drive AI innovation with minimal regulatory friction.

2. Background on Lexset

Lexset is a leader in synthetic data development, harnessing advanced generative techniques to create high-fidelity, photorealistic datasets essential for training state-of-the-art AI models. Founded in 2018 and based in Brooklyn, NY and Gig Harbor, WA, Lexset is backed by top-tier investors such as Super Ventures and XRC Labs, and has earned recognition as a Verizon "Built on 5G Challenge" winner.

Our flagship platform, Seahaven, revolutionizes synthetic data generation by combining graph-based scene creation, procedural simulation, and robust physics-based controls. This technology enables the rapid production of diverse training data across multiple sensor modalities-from Synthetic Aperture Radar (SAR) and Radar to EO imagery-addressing critical data shortages that hamper traditional AI development. Lexset's approach not only accelerates the training process but has been shown to improve object detection accuracy by nearly threefold, making it indispensable for both commercial and defense applications.

Lexset has successfully partnered with leading technology companies and manufacturing companies, as well as U.S. government agencies including the U.S. Air Force, and the U.S. Space Force. Through partnerships - such as our engagements via the National Security Innovation Network (NSIN) and various DoD Small Business programs - we have demonstrated how synthetic data can enhance autonomous systems, improve intelligence analysis, and support advanced operational scenarios. These collaborations have enabled rapid prototyping and deployment of AI solutions that drive down costs and reduce the need for extensive, real-world data collection, supporting government spending efficiency.

By integrating cutting-edge synthetic data methodologies with strong industry partnerships, Lexset is uniquely positioned to support the U.S. government's strategic objectives of sustaining AI leadership, ensuring national security, and maintaining a competitive edge against global rivals. Our commitment to innovation and collaboration provides a powerful model for how public-private partnerships can transform AI R&D, accelerate technology transfer, and deliver robust, cost-effective solutions for defense and national security applications.

2. The Role of Synthetic Data in AI Development for Defense and National Security

Synthetic data has emerged as a critical tool for advancing AI. This is especially true in domains where real data is scarce, unstructured and unwieldy, or costly to obtain. In defense and national security applications, synthetic data enables the training and testing of AI models for scenarios that are difficult or dangerous to replicate. It can augment or substitute for real-world data when the latter is unavailable or exhausted.[1] For example, the U.S. Army has identified that many AI/ML models in the defense domain suffer from a "bottleneck" of training data; an Army program noted there is "not enough unique defense/intel data available to create highly performant models", and that synthetic data generation could alleviate this shortage.[2] By generating realistic sensor feeds, images, or simulations of operational environments, synthetic data allows AI systems to learn to recognize threats (e.g. vehicles, targets, or anomalies) without requiring massive real world datasets.

These capabilities are already at work. In a collaboration with the Air Force Research Lab, Lexset's photorealistic synthetic imagery was used to train computer vision algorithms for unmanned systems, boosting object detection accuracy and overall autonomous system performance. This demonstrates how simulated data of battlefields or surveillance scenes can sharpen AI perception for warfighters. Similarly, the Department of Homeland Security has recognized synthetic data as a *"gamechanger"* that allows training AI when real data is too sensitive or not available.[3] DHS's 2024 solicitation for synthetic data solutions highlights that synthetic data enables model training while safeguarding privacy and security, particularly for scenarios involving personally identifiable information. Across the national security community, there is growing consensus that synthetic data generation through simulation and modeling is a critical tool to create the large, varied datasets needed for robust AI applications.[4]

3. Cost-Saving Benefits of Synthetic Data and Efficient AI Development

Synthetic data can significantly reduce the cost and time of AI development, acting as a force multiplier for government and industry resources. Traditional AI model training often requires collecting and using human laborers to annotate millions of real-world data points, an expensive and labor-intensive process. In contrast, synthetic datasets are generated in virtual environments with built-in ground truth labels, eliminating much of the manual labeling cost. The

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[1] House Artificial Intelligence Task Force Final Report (2024)
[2] U.S. Army SBIR Topic A214-042: Sensor Synthetic Data Generation (2021)
[3] OHS S&T Announces New Solicitation for Synthetic Data Generator Solutions (2024)
[4] National Security Commission on AI Final Report (2021)

U.S. Army notes that creating a moderately accurate model might require on the order of 50 million real data samples, whereas a synthetic data pipeline can produce unlimited labeled examples on demand.[5] By leveraging synthetic data, agencies can avoid costly data collection campaigns and rapidly produce specialized datasets (e.g. satellite images of missile launchers or radar signals of targets) tailored to their mission needs. There is no need to wait months or years to accumulate real-world examples of every scenario. This on-demand data generation not only saves money but also accelerates development cycles.

Moreover, synthetic data enables extensive pre-testing of AI systems before real-world deployment, catching potential issues early and cheaply. Models can be trialed against countless simulated edge cases (e.g., rare conditions, adversarial tactics, and extreme environments) that would be logistically difficult or expensive to reproduce in live exercises. This comprehensive virtual testing improves reliability and safety, reducing the risk of costly failures when new systems are fielded.

In the defense context, battlefield simulations can ensure AI-driven systems are robust against a wide range of threat scenarios, thereby avoiding expensive fixes or mission failures later. Industry analyses reinforce these benefits: synthetic data brings cost efficiency by cutting expenses associated with real data collection, labeling, and yields time savings by accelerating AI training and testing pipelines.

In short, every dollar invested in high-quality synthetic data can return many dollars in saved labor, faster development, and more reliable AI outcomes. Harnessing synthetic data thus supports government spending efficiency. Agencies retooling for future challenges like the Department of Defense and the Department of Homeland Security can do more with constrained budgets, focusing funds on critical hardware or personnel needs rather than on protracted data-gathering efforts.

4. Strategic Importance of Synthetic Data in Maintaining U.S. AI Leadership

Maintaining America's lead in artificial intelligence is not just an economic issue but a national security imperative, especially in light of strategic competition with China. The People's Republic of China has declared its ambition to become the world leader in AI by 2030, backing this goal with massive investments and state-driven AI programs. Chinese researchers have already surpassed their U.S. counterparts in sheer volume of AI research publications in recent years, and China's private sector and government spending on AI are growing rapidly.[6] U.S. policy, as articulated in EO 14179, is well-directed to "sustain and enhance America's global AI dominance" for the sake of economic and national security.[7] In this context, synthetic data is a strategic asset that can help the United States outpace competitors in AI development.
China's advantages in AI often stem from its access to vast amounts of data and a centrally coordinated national AI strategy. The U.S. can counter these advantages by leveraging

[5] U.S. Army SBIR Topic A214-042: Sensor Synthetic Data Generation (2021)
[6] Congressional Research Service, AI: Background, Issues, and Policy (R46795, 2021)
[7] Executive Order 14179, Removing Barriers to American Leadership in Artificial Intelligence (2025)

innovation in synthetic data to overcome our own data limitations. By generating unlimited training data virtually, American researchers and companies can *overcome the* challenges in obtaining real data, leveling the playing field or even leapfrogging in our competitor's capabilities. If the U.S. leads in synthetic data generation technology, it will lead in the ability to train the most advanced AI systems at scale.

This has direct national security implications: superior AI trained on richer, more diverse data will enhance U.S. military intelligence, autonomous platforms, and decision-support systems relative to those of other nations. Conversely, if the U.S. falls behind, we risk a world in which Chinese standards and technologies dominate, a concern expressed by Congress and defense experts who note that Beijing is aggressively pursuing leadership in AI and even international AI standards-setting.[8] American leadership in synthetic data and AI is thus essential to ensure that the values of openness, privacy, and human rights prevail over authoritarian models of AI use. Synthetic data is strategic infrastructure for AI superiority. By investing in it, the United States can accelerate innovation, maintain a military-technological edge, and blunt China's attempts to gain AI dominance.

5. Policy Recommendations for Advancing AI Innovation

In line with Executive Order 14179's objectives, Lexset offers the following recommendations to support U.S. AI leadership and innovation. These recommendations aim to foster AI development, particularly for defense applications, while minimizing regulatory burdens, thus enabling the U.S. to maintain its edge in an increasingly competitive global AI landscape:

I. **Promote Synthetic Data Adoption Across Federal AI Initiatives.** Federal agencies should explicitly encourage and fund the use of synthetic data in AI R&D, testing, and evaluation. This could include directing R&D funding (via NSF, DoD, OHS S&T, etc.) toward synthetic data generation tools and establishing pilot programs where agencies partner with industry to create synthetic datasets for high-priority AI challenges. By integrating synthetic data into programs like the DoD Joint AI Center initiatives, the DoD Data Strategy, and NSF-funded research, the government can amplify AI progress with limited real-world data. Agencies should develop standards and best practices for validation of synthetic data quality, as highlighted by the DOD's National Science and Technology Council's 2023 strategy calling for research into verification techniques for synthetic data accuracy and effectiveness.[9] Ensuring synthetic data is rigorously evaluated will build trust in its use. We also recommend establishing a public synthetic data catalog or sandbox (perhaps as part of the National AI Research Resource) where researchers and companies can access government-curated synthetic datasets (with no privacy issues) for AI development. This would accelerate innovation across academia, industry, and government by making *"open synthetic data"* a shared resource.

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[8] Congressional Research Service, AI: Background, Issues, and Policy (R46795, 2021)
[9] National Defense Science & Technology Strategy (2023)

II. **Strengthen Opportunities for Small Businesses in Defense AI Innovation.** The U.S. government should expand mechanisms that allow small innovative AI companies to contribute to national security missions. Lexset's experience through small business contracts and the NSIN accelerator suggests that relatively small investments in nimble businesses can yield outsized benefits by bringing cutting-edge tech into government use. We urge the continuation and growth of programs like AFWERX, DIU (Defense Innovation Unit), NSIN, and DHS's Silicon Valley Innovation Program, which connect non-traditional vendors/startups with defense and security needs. Contracting processes should be further streamlined to lower barriers for small AI firms to work with the government (e.g., faster acquisition pathways, flexible intellectual property arrangements, and bridge funding between pilot projects and full adoption). By lowering contracting and compliance hurdles, the government can tap into the full creativity of the private sector. We also recommend incentives for large prime contractors to incorporate innovative subcontractors in defense programs, fostering a collaborative ecosystem. Ultimately, robust public-private collaboration will help ensure the U.S. military and homeland security agencies have access to the best AI technologies available domestically, reinforcing our technological and military dominance.

III. **Leverage Synthetic Data to Improve Government Efficiency and Reduce Costs.** As agencies implement AI solutions, they should prioritize cost-effective strategies like synthetic data pre-testing. We recommend 0MB and agency leadership issue guidance recognizing synthetic data as a best practice for AI project planning - for instance, requiring that proposals for new AI systems consider the use of synthetic data during development to optimize cost and performance. By conducting extensive virtual experimentation, agencies can identify failures or biases in AI models without the expense of live trials, thus avoiding wasteful spending. Successful case studies (such as the Army's use of synthetic data to overcome intel data gaps)[10] should be shared across government to illustrate return on investment. Making synthetic data a cornerstone of federal AI development will maximize the impact per taxpayer dollar, enabling high-performance AI systems without the traditionally high costs of data acquisition and model training.

IV. **Maintain Light-Touch Regulation and Remove Barriers to AI Deployment.** We strongly support the intent of EO 14179 to eliminate or revise policies that needlessly hinder AI innovation. To this end, we recommend reviewing and rescinding legacy regulations or agency policies that inadvertently discourage data sharing or AI adoption. The federal government should also coordinate with state and local governments to discourage a patchwork of AI regulations; a clear national standard will help American AI companies innovate with confidence. In international contexts, U.S. diplomats and trade officials should push back against foreign regulations that unfairly disadvantage U.S. AI products or that seek to impose another country's values via technology standards. By championing a pro-innovation regulatory climate at home and abroad, the U.S. can

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[10] U.S. Army SBIR Topic A214-042: Sensor Synthetic Data Generation (2021)

ensure that its AI developers (large and small) are not hamstrung by red tape, and that American AI systems reach the field faster to compete globally. We also endorse continued inter-agency efforts to clarify ethical guidelines and risk management practices for AI (building on NIST's work), so that essential guardrails are in place without imposing heavy-handed rules that deter experimentation.

V. **Prioritize National Security AI R&D and Leadership Coordination.** To secure long-term U.S. leadership, federal policy should treat AI, including synthetic data capabilities, as a strategic priority on par with other critical technologies. We applaud the inclusion of national security considerations in the AI Action Plan mandate, and we urge sustained focus on defense-related AI in its implementation. Concretely, this means increasing investments in AI R&D for defense and dual-use technologies. It also means cultivating the AI talent and workforce needed for the defense sector, possibly by expanding scholarships and career pathways for AI engineers to enter public service or defense industry roles. Another aspect of leadership is international: the U.S. should lead alliances for AI cooperation - sharing data (including synthetic data techniques) and aligning AI ethics with allies - to magnify our collective capabilities against adversaries. Finally, high-level coordination is key. We recommend empowering the White House's science and technology leaders and the *National Security Advisor* (as named in EO 14179) to continually identify and remove any emerging "barriers to American AI leadership." This could involve quick action to address bottlenecks such as export controls on certain AI hardware, immigration hurdles for AI talent, or insufficient tech transfer out of federal labs. By proactively tackling such issues, the Administration can ensure nothing stands in the way of U.S. technological and military dominance in AI. A whole-of-government commitment to AI leadership - with defense innovation at its core - will signal to both allies and competitors that the United States intends to retain its preeminent position in AI for the foreseeable future.

Lexset appreciates the opportunity to contribute to the formulation of the U.S. Artificial Intelligence Action Plan. Synthetic data is a powerful enabler of AI advancement, and with thoughtful policy support it can help drive breakthroughs in defense, security, and many civilian sectors while saving costs and safeguarding privacy. The United States stands at a pivotal moment. By embracing innovation and removing impediments, we can unleash AI's potential to strengthen our economy and secure our nation. In drafting the Action Plan, we urge NSF, OSTP, and NITRO to incorporate the above recommendations, focusing on data-driven approaches, strategic investment, and light-touch regulation, which align with the mandate of Executive Order 14179 to sustain American AI leadership. Lexset remains committed to partnering with the U.S. government to achieve these goals. We are prepared to share further insights from our R&D efforts and to support initiatives that will ensure the United States continues to lead in artificial intelligence against any competition. Together, government and industry can create an environment where AI innovation flourishes on American soil, bolstering our national security and prosperity for years to come.